IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

MATHEMATICS FOR SIGNALS AND SYSTEMS

Wednesday, 3 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Dragotti
Second Marker(s) : S. Parbhoo

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Special Information for the Invigilators: none

Information for Candidates:

For questions involving matrix manipulations or calculations, please provide answers showing intermediate calculations steps.

The function sign(a) is define as follows:

$$\text{sign}(a) = \begin{cases} 1 & \text{for } a \geq 0 \\ -1 & \text{for } a < 0 \end{cases}$$

The inverse of a non-singular matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \mathbf{C}^T$$

where the entry i, j of $\mathbf{C}$ is the co-factor $c_{i,j}$ given by $(-1)^{i+j}\mathbf{M}_{i,j}$ where the 'minor' $\mathbf{M}_{i,j}$ is defined to be the determinant of the $(n-1) \times (n-1)$-matrix that results from $\mathbf{A}$ by removing the i-th row and the j-th column.

The norm of a vector $\mathbf{x}$ is $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T\mathbf{x}}$.

The Questions

1. (a) Construct a matrix with the characteristics described below. If what is required is not possible, explain why. Please note that the solution might not be unique.

   i. Construct a matrix where the column space has dimension two and contains $[1, 1, 0]^T$, moreover the row space contains $[1, 2]^T$ and $[2, 5]^T$. [2]

   ii. Construct a matrix where the left null space contains $[1, 3]^T$ and the row space contains $[3, 1]^T$. [2]

   iii. Construct a matrix where the column space contains $[1, -1, 1]^T$ and $[2, 1, 3]^T$ and the null space contains $[1, -1]^T$. [2]

   (b) Suppose $\mathbf{W} \in \mathbb{R}^{m \times m}$ and $\mathbf{Q} \in \mathbb{R}^{n \times n}$ are two unitary matrices.

   i. Show that $\mathbf{A}$ and $\mathbf{C} = \mathbf{WAQ}$ have the same singular values. [2]

   ii. Compute the singular values and the singular vectors of:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}.$$

[2]

   iii. Compute the singular values and the singular vectors of:

$$\mathbf{C} = \begin{bmatrix} 1 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

[2]

Question 1 continues on the next page

(c) Given

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0.01 \end{bmatrix} \text{ and } \mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

i. find the solution to the problem:

$$\mathbf{A}\mathbf{x}_1 = \mathbf{b}_1.$$

[1]

ii. Assume now that the true $\mathbf{b}_1$ has been corrupted by noise. Find the solution to

$$\mathbf{A}\mathbf{x}_2 = \mathbf{b}_2,$$

with $\mathbf{b}_2 = \mathbf{b}_1 + \mathbf{e} = [1.1, 1.2]^T$ and verify that

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|}{\|\mathbf{x_1}\|} > \frac{\|\mathbf{b}_1 - \mathbf{b}_2\|}{\|\mathbf{b}_1\|}.$$

[1]

iii. To increase resiliency to noise, instead of solving $\mathbf{A}\mathbf{x} = \mathbf{b}$ directly, solve the regularized version of your problem:

$$\min_x \left(\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2 \right).$$

Show that the solution is equivalent to $\mathbf{x}_\lambda = (\mathbf{A}^T\mathbf{A} + \lambda^2\mathbf{I})^{-1}\mathbf{A}^T\mathbf{b}$. [3]

iv. Find λ so that

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_\lambda\|}{\|\mathbf{x_1}\|} < 5\frac{\|\mathbf{b}_1 - \mathbf{b}_2\|}{\|\mathbf{b}_1\|}.$$

[3]

2. (a) Mark each statement below true or false and briefly justify you answer. Let V be a set of m vectors in $\mathbb{R}^n$.

   i. If $n > m$ then the vectors in V are linearly independent. [1]

   ii. If the zero vector is in V then the vectors in V are linearly dependent. [1]

   iii. If the vectors in V are linearly independent and T is a subset of V, then the vectors in T are linearly independent. [1]

   iv. If the vectors in T are linearly dependent and T is a subset of V, then the vectors in V are linearly dependent. [1]

   (b) Consider the following set of four measurements:

$$(t_i, y_i) : (1, 0.5), (2, 1), (3, 1.5), (4, 2).$$

   i. Find the linear polynomial $p(t) = a_0 + a_1 t$ that best fits these points in the least-squares sense. [3]

   ii. Determine now the linear least-squares fit to the data, under the strict constraint that the polynomial satisfies $a_0 + a_1 t = 3.25$ for $t = 5$. [5]

   (c) Consider the symmetric Markov or "transition" matrix $\mathbf{Q} \in \mathbb{R}^{3\times 3}$. You are told that the probability of going from state one to state two is $p_{2,1} = 0.125$ and from state one to state three is $p_{3,1} = 0.25$. You are also told that one eigenvalue of $\mathbf{Q}$ is $\lambda = 0.5$ and that $\det(\mathbf{Q}) = 0.125$.

   i. Find $\mathbf{Q}$. [4]

   ii. Assume $\mathbf{Q}$ models the fraction of population that moves every year across three different towns. Find the steady state of $\mathbf{Q}$. [2]

   iii. Assume the total population at the beginning of your observation is $90,000$. What is the population in each town at the steady state? [2]

3. (a) Given

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix} \text{ and } \mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

find minimum norm solution to this problem.

[4]

(b) A Householder matrix is defined by

$$\mathbf{H}(\mathbf{v}) = \mathbf{I} - 2\frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}, \quad \mathbf{v} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$$

i. Show that $\mathbf{H}(\mathbf{v})$ is orthogonal, specifically show that

$$\mathbf{H}(\mathbf{v})^T\mathbf{H}(\mathbf{v}) = \mathbf{I}.$$

[4]

ii. Let $\mathbf{a} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$ be a nonzero vector. Denote with a_1 the first entry of $\mathbf{a}$ and let $\mathbf{e}_1 = [1, 0, \ldots, 0]^T$. Define the scalars

$$s = \text{sign}(a_1) \;\; \text{(with the convention } \; \text{sign}(0) = 1),$$

and

$$\alpha = s\,\|\mathbf{a}\|$$

with $\|\mathbf{a}\| = \sqrt{\mathbf{a}^T\mathbf{a}}$. Moreover, let $\mathbf{v} = \mathbf{a} + \alpha\mathbf{e}_1$.
Show that

$$\mathbf{H}(\mathbf{v})\,\mathbf{a} \;=\; -\,s\,\|\mathbf{a}\|\,\mathbf{e}_1$$

with

$$\mathbf{H}(\mathbf{v}) \;=\; \mathbf{I} \;-\; 2\,\frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}.$$

[4]

Question 3 continues on the next page

(c) The QR factorization of a matrix $\mathbf{A}$ can be achieved using Householder matrices.

i. Assume $\mathbf{A} \in \mathbb{R}^{n \times n}$ is non-singular. Construct $\mathbf{H}_1 \in \mathbb{R}^{n \times n}$ that zeroes the entries below a_{11} in the first column of $\mathbf{A}$. Specifically, construct $\mathbf{H}_1$ so that

$$\mathbf{A}^{(1)} = \mathbf{H}_1 \mathbf{A} = \begin{bmatrix} \alpha & \mathbf{r}_{12}^T \\ \mathbf{0} & \mathbf{A}_{22} \end{bmatrix},$$

with $\alpha = \pm \|\mathbf{a}_1\|$, $\mathbf{r}_{12}^T \in \mathbb{R}^{1 \times (n-1)}$ and $\mathbf{A}_{22} \in \mathbb{R}^{(n-1) \times (n-1)}$. Here $\mathbf{a}_1$ is the first column of $\mathbf{A}$. [Hint: use the findings of part (b)] [4]

ii. Given

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix},$$

find $\mathbf{H}_1$ so that the entries of $\mathbf{A}$ below a_{11} are zero. [4]